

Bio Fusion

Solve an industry-relevant AI in Medical Diagnostics Challenge
2026

Project Submission Report

Automated Grading of Gastric Cancer Histopathology using Vision Transformers

A Deep Learning Approach to Clinical Decision Support

Team Name: InsightAI

Team Members:

Name	Role
Oshada Kariyawasam	Team Leader
Sasindu Wijewardana	Member-1
Kavinu Saputhanthri	Member-2

January 4, 2026

Project Title:

Gastric Cancer Histopathology Tissue Classification using Hybrid Ensemble (EfficientNet + ViT) and Baseline ConvNeXt

1. Problem Identification and Significance

1.1 Clinical Context and Stakeholders

Gastric (stomach) cancer is among the leading causes of cancer-related mortality worldwide. Despite advances in oncology, the diagnostic workflow largely relies on the manual examination of Hematoxylin and Eosin (H&E) stained slides by pathologists. This traditional workflow faces significant bottlenecks. Pathologists are burdened by increasing case volumes and time pressures, making detailed quantitative assessment of every slide difficult to scale. The primary stakeholders affected include pathologists, who require decision-support tools to reduce fatigue; hospital laboratories, which need standardization and Quality Assurance (QA); and ultimately patients, for whom diagnostic delays or variability can adversely affect treatment decisions.

1.2 The Importance of the Tumor Microenvironment (TME)

The clinical course of gastric cancer is strongly influenced by the Tumor Microenvironment (TME)—the ecosystem surrounding the tumor cells. The ratio of tumor to stroma (Tumor-Stroma Ratio or TSR) and the presence of immune infiltrates (lymphocytes) are critical biomarkers for prognosis. However, quantifying these components visually is subjective and prone to high inter-observer variability.

1.3 Unmet Healthcare Need

There is a specific, unmet need for **consistent, automated tissue component recognition** on H&E slides. Current manual methods cannot reliably quantify complex TME metrics at scale. An automated solution would reduce variability, accelerate triage, and enable the extraction of downstream biomarkers (e.g., TSR, immune hotspots) that are currently too labor-intensive to measure routinely. This project addresses this gap by developing a classifier capable of granular tissue decomposition.

2. Dataset Justification

2.1 Dataset Source and Relevance

The study utilizes the **HMU-GC-HE-30K (GCHTID)** dataset, derived from 300 H&E-stained gastric cancer Whole Slide Images (WSIs) scanned at 20× magnification. The dataset was tiled into 224×224 pixel patches and labeled through a multi-reader pathology workflow. This dataset is uniquely appropriate because it was designed specifically for multi-class TME analysis rather than simple binary (cancer/non-cancer) detection.

2.2 Alignment with Clinical Goals

The dataset labels align directly with the project’s objective to decompose tissue into functional components. The eight classes are:

- **ADI:** Adipose tissue
- **DEB:** Debris/Necrosis
- **LYM:** Lymphocyte aggregates (immune response)
- **MUC:** Mucus
- **MUS:** Smooth muscle
- **NOR:** Normal gastric mucosa
- **STR:** Stroma (connective tissue)
- **TUM:** Tumor epithelium

This taxonomy allows for the calculation of clinically vital metrics, such as the ratio of TUM to STR or the density of LYM patches.

2.3 Data Volume and Balance

With approximately 31,000 patches (roughly 3,887 per class), the dataset is exceptionally balanced. In medical imaging, class imbalance often biases models toward the majority class (e.g., healthy tissue). The balanced nature of HMU-GC-HE-30K supports stable multi-class training and ensures that metrics like Macro-F1 are reliable indicators of performance across all tissue types.

3. Literature Review and Innovation

3.1 Historical Limitations in Computational Pathology

Early datasets for gastric histopathology, such as GasHisSDB, focused primarily on binary classification. While useful for basic benchmarking, these datasets fail to capture the complexity of the TME. A binary model can flag a region as "abnormal," but it cannot distinguish between a high-stroma tumor (poor prognosis) and a lymphocyte-rich tumor (potentially better prognosis). This represents a significant gap in the literature regarding granular tissue phenotyping.

3.2 Evolution of Backbones: CNNs vs. Transformers

The field of computational pathology has evolved from standard Convolutional Neural Networks (CNNs) to modern architectures:

- **EfficientNet** introduced compound scaling to optimize accuracy and efficiency, excelling at capturing local textures and edges.
- **Vision Transformers (ViT)** shifted the paradigm by processing images as sequences of patches, allowing the model to capture global context and long-range dependencies.
- **ConvNeXt** modernized the CNN architecture by adopting Transformer-like design choices (e.g., larger kernels, layer normalization) while retaining the inductive bias of convolutions.

3.3 The Gap and Proposed Solution

A critical observation in histopathology is that no single architectural inductive bias dominates. CNNs are superior at identifying local morphological textures (e.g., nuclear atypia), while Transformers excel at understanding the global arrangement of tissue structures.

Innovation: This project proposes a Hybrid Ensemble that fuses EfficientNet-B4 and ViT-Base. Unlike standard ensembles that average predictions, this approach uses a learned attention module to weight the feature vectors from each backbone dynamically. This aims to resolve class confusions among morphologically similar tissues (e.g., Stroma vs. Tumor) by leveraging both local and global signals.

4. Methodology

4.1 Data Preprocessing Pipeline

To ensure robustness against stain variability—a common issue in histopathology—a rigorous preprocessing pipeline was implemented:

1. **Label Mapping:** Organized data into a folder-per-class structure corresponding to the 8 TME classes.
2. **Stratified Splitting:** Data was split into Train (~70%), Validation (~15%), and Test (~15%) sets, preserving the class distribution to prevent sampling bias.

3. **Contrast Enhancement (CLAHE):** Contrast Limited Adaptive Histogram Equalization was applied. Unlike global equalization, CLAHE operates on small regions, enhancing local contrast and defining tissue boundaries without amplifying noise.
4. **Augmentation (Train Only):** To prevent overfitting, images underwent random horizontal/vertical flips, small rotations ($\pm 15^\circ$), and mild color jittering. Aggressive color shifts were avoided to maintain histological realism.
5. **Normalization:** Images were normalized using ImageNet mean/std statistics to align with the pretrained backbones.

4.2 Model Architectures

A) Baseline: ConvNeXt-Base

We selected ConvNeXt-Base as the single-model baseline. It represents the state-of-the-art in pure CNN design, utilizing depth-wise separable convolutions and an inverted bottleneck design. It serves as a rigorous benchmark to determine if the complexity of a hybrid model is justified.

B) Hybrid Ensemble (EfficientNet-B4 + ViT-Base)

The proposed architecture runs two parallel backbones:

1. **EfficientNet-B4:** Extracts a feature vector rich in local textural information.
2. **ViT-Base:** Extracts the classification ([CLS]) token, rich in global context.

Fusion Mechanism:

Instead of simple concatenation, an Attention MLP (Multi-Layer Perceptron) receives the concatenated features and outputs two scalar weights via a Softmax function.

This fused vector is passed to a classifier head containing Dropout regularization and fully connected layers to produce the final 8-class logits.

4.3 Training Strategy

A **Two-Phase Training** strategy was employed to maximize transfer learning efficiency while preserving pretrained features:

- **Phase 1 (Feature Extraction):** Backbones were frozen. Only the fusion module and classifier head were trained (Learning Rate: $1e-3$). This aligns the random weights of the head with the pretrained features.
- **Phase 2 (Fine-Tuning):** The top blocks of both backbones were unfrozen and fine-tuned with a lower learning rate ($1e-4$) using the AdamW optimizer and a Cosine Warm Restart scheduler. This allows the model to adapt specifically to gastric histology without catastrophic forgetting of ImageNet features.

5. Results and Discussion

Metric	ConvNeXt-Base (Baseline)	Hybrid Ensemble (EfficientNet+ViT)	Difference
Test Accuracy	73.05%	69.43%	-3.62%
Macro F1	0.7339	0.6940	-0.0399
MCC	0.6931	0.6508	-0.0423

5.1 Why did ConvNeXt win overall?

The "pure" CNN (ConvNeXt) outperformed the Hybrid model in general accuracy (73.05% vs 69.43%).

- **Texture Dominance:** Histopathology is often texture-dominant. ConvNeXt is optimized for hierarchical feature extraction, which may be more robust for distinguishing dense tissues like Muscle (MUS) and Stroma (STR).
- **Optimization Difficulty:** The Hybrid model requires optimizing two massive backbones plus a fusion module. The lower performance suggests the ensemble may have suffered from optimization difficulties or slight overfitting, as evidenced by the complexity not yielding a "sum greater than its parts."

5.2 The "Silver Lining": Superior Immune Detection

Crucially, the **Hybrid Ensemble significantly outperformed the Baseline in Lymphocyte (LYM) Recall (85.1% vs 76.2%)**.

- **Interpretation:** Lymphocytes appear as scattered, small dark nuclei. Detecting them often requires "global context" (counting density across the patch) rather than just local texture. This validates the hypothesis that the **ViT component** of the ensemble contributes valuable global attention, making the model more sensitive to immune infiltrates even if it loses specificity elsewhere.

5.3 Critical Confusion: Tumor vs. Stroma

Both models struggled with the **STR vs. TUM** distinction, but the Hybrid model struggled more (STR F1: 0.629 vs Baseline: 0.680). This separation is the most clinically vital for Tumor-Stroma Ratio calculation. The Hybrid model's attention mechanism seemingly failed to disentangle the complex interface where tumor cells invade stromal tissue.

6. Conclusion and Future Work

6.1 Conclusion

This study benchmarks a modern ConvNet against a Hybrid Transformer-CNN ensemble for gastric TME classification. While the **ConvNeXt-Base** proved to be the more robust and accurate generalist (73.05%), the **Hybrid Ensemble** demonstrated a specific advantage in detecting immune response (Lymphocytes), achieving 85.1% recall. This suggests that while ConvNets are sufficient for structural segmentation, Transformer-based fusion is promising for detecting scattered cellular patterns.

6.2 Recommendations for Improvement

To bridge the gap and leverage the Hybrid model's potential:

1. **Refine Fusion Strategy:** Replace the simple Attention MLP with a **Cross-Attention** mechanism (where CNN features query ViT keys/values) to better align local and global features spatially.
2. **Hard Negative Mining:** Specifically retrain the model on STR/TUM and NOR/MUC confusion pairs, as these drive the majority of errors in both architectures.
3. **Ensemble via Voting:** Rather than feature fusion, a "Soft Voting" ensemble of the trained ConvNeXt and the ViT might yield better results than end-to-end feature training, combining ConvNeXt's precision with ViT's recall.

7 Real-world Application and Marketing

7.1 Deployment Scenario

The proposed model is designed to function as a **Digital Pathology Assistant**. In a clinical setting, the workflow would involve:

1. Ingesting a WSI from a digital scanner.
2. Tiling the WSI into patches.
3. Generating a color-coded heatmap overlaid on the slide, highlighting regions of Tumor, Stroma, and Immune infiltration.
4. Automatically calculating the Tumor-Stroma Ratio (TSR), a prognostic factor that is tedious to calculate manually.

7.2 Market Impact

The target market includes mid-to-large hospital networks and research laboratories.

- **Benefit:** Reduces the time pathologists spend on routine tissue identification, allowing them to focus on complex diagnostic decisions.
- **Cost-Effectiveness:** Designed as a software-only add-on compatible with existing viewers (e.g., QuPath), avoiding the need for expensive proprietary hardware.
- **Research Value:** Provides reproducible data for clinical trials that require stratification of patients based on immune phenotype.

8. Future Improvements and Conclusion

8.1 Future Enhancements

To move from prototype to clinical deployment, several improvements are planned:

- **Stain Normalization:** Implementing Macenko normalization to mathematically align color spaces across different scanners, further reducing domain shift risks.
- **Discriminative Learning Rates:** Applying lower learning rates to the pretrained backbone layers and higher rates to the new heads during Phase 2 to improve stability.
- **Hard Negative Mining:** Specifically targeting the NOR/MUC and STR/TUM confusion pairs by oversampling these difficult examples during training.

8.2 Conclusion

This project demonstrates the feasibility of detailed, 8-class gastric tissue phenotyping using deep learning. The ConvNeXt baseline established a strong foundation with 73% accuracy, identifying key challenges in distinguishing distinct stromal and epithelial interfaces. The proposed Hybrid Ensemble addresses these limitations by fusing the textural precision of CNNs with the contextual awareness of Transformers. By automating the quantification of the Tumor Microenvironment, this tool holds the potential to standardize histopathology reporting and unlock new prognostic insights for gastric cancer patients.

References

- [1] International Agency for Research on Cancer. (2020). *Stomach Cancer Fact Sheet*. <https://gco.iarc.fr/today/data/factsheets/cancers/7-Stomach-fact-sheet.pdf>
- [2] Wang, X., et al. (2024). A large histological images dataset of gastric cancer with pixel-level annotations. *Nature Scientific Data*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11754904/>
- [7] Liu, Z., et al. (2022). A ConvNet for the 2020s. *CVPR 2022*. <https://arxiv.org/pdf/2201.03545.pdf>

Git Hub Repo: <https://github.com/Oxshadha/Bio-Fusion.git>